House Prices Analysis

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Abstract

This study focuses on modeling the price of a bunch of houses using a Generalized Linear Model (GLM) with a Gamma distribution to account for the positively skewed nature of the target variable. The analysis will show the impact that certain variables present within the dataset have on housing prices. In fact, We'll see that not all of them will prove significant according to the model chosen for the study. Let's see, first of all, a brief introduction regarding the variables.

1. Introduction

The problem

The dataset, available on www.kaggle.com, contains 414 different houses and 8 variables.
The following are the last ones:

- Age: a continous variable that shows how old is the house from the date of make.
- Nearest Metro: shows the distance in km from nearest metro station.
- Convenience stores: a counting variable that represents the number of nearby stores available near house.
- latitude: The latitude of the house, measured in degrees.
- longitude: The longitute, like the latitude, in degrees.
- Bedrooms: A count regarding the number of bedrooms into the house.
- House size: Total area of the house in sqft.
- Price: The price of the house per unit area, expressed in dollar.

Exploratory Data Analysis

We start with a summary of the data to gain a general understanding of them structure, including the distributions of features, data types, and potential anomalies.

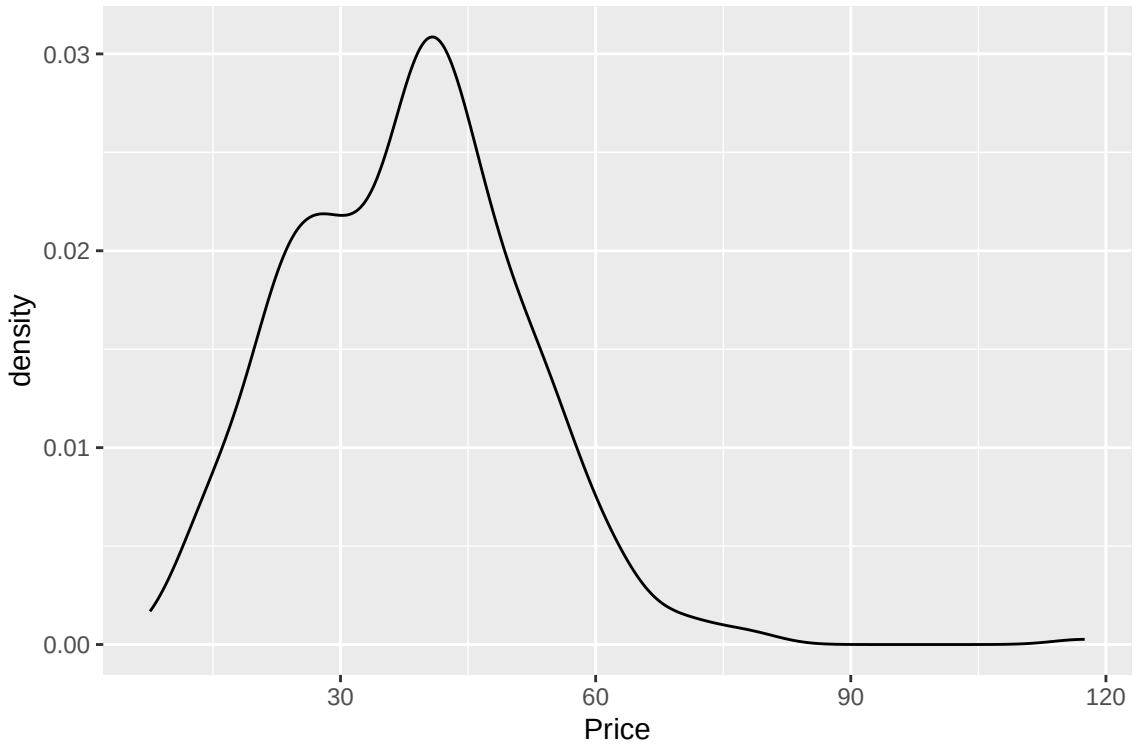
```
##       Age         Nearest metro     Convenience stores    latitude
##  Min.   : 0.000   Min.   :  23.38   Min.   : 0.000     Min.   :24.93
##  1st Qu.: 9.025   1st Qu.: 289.32   1st Qu.: 1.000     1st Qu.:24.96
##  Median :16.100   Median : 492.23   Median : 4.000     Median :24.97
##  Mean   :17.713   Mean   :1083.89   Mean   : 4.094     Mean   :24.97
##  3rd Qu.:28.150   3rd Qu.:1454.28   3rd Qu.: 6.000     3rd Qu.:24.98
##  Max.   :43.800   Max.   :6488.02   Max.   :10.000     Max.   :25.01
##    longitude        Bedrooms       House size         Price
```

```
##  Min.   :121.5   Min.   :1.000   Min.   : 402.0   Min.   :  7.60
##  1st Qu.:121.5   1st Qu.:1.000   1st Qu.: 548.0   1st Qu.: 27.70
##  Median :121.5   Median :2.000   Median : 975.0   Median : 38.45
##  Mean   :121.5   Mean   :1.988   Mean   : 931.5   Mean   : 37.98
##  3rd Qu.:121.5   3rd Qu.:3.000   3rd Qu.:1234.8   3rd Qu.: 46.60
##  Max.   :121.6   Max.   :3.000   Max.   :1500.0   Max.   :117.50
```

This summary provides an overview of the dataset's variables and their key statistics:

- age: the variable related to the age of the house has a range from 0 to about 44 years old, with with an average age of about 17 and a half years.
- nearest metro: on average, the distance to the nearest metro station is about 1000 km. Looking at the median, we notice a strong right skew, as half of the houses are located up to 300 km away, while the farthest house from the metro is about 6500 km away.
- convenience stores: the variable counts the number of nearby convenience stores from 0 to 10; the houses have an average of 4 nearby convenience stores.
- latitude: The range of latitude is quite narrow, which indicates that it is a small urban area. The mean and the median are both equal to 24.97.
- longitude: together with the latitude represents the coordinates of the area in question; it seems to be a small town in Taiwan. Here too, the range is very small and the mean and median values coincide.
- Bedrooms: On average, the houses in the analysis have 2 rooms (in a count from 1 to 3).
- House size: measured in sqft, house sizes have a range from 400 to 1500; values of mean and median are both near to 950, that shows a distribution fairly balanced.
- Price: The price of houses per unit area represents the target variable. The mean and median values of about 38, on a range from 7.6 to about 120, show the evident right skewness that characterizes this variable.

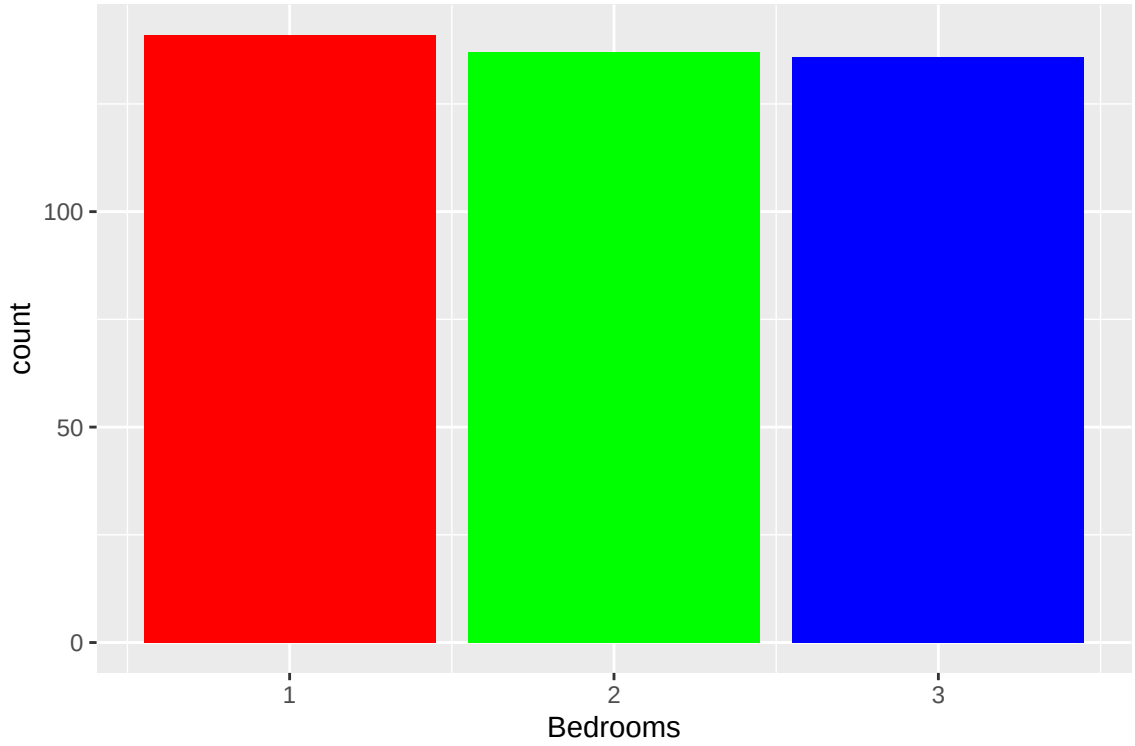
Now, let's analyze the target variable, by visualizing its density plot to explore its distribution and how the features relate to it. This analysis will help uncover potential patterns, relationships and key factors influencing the target variable. By visualizing and summarizing the data, we aim to identify the most significant predictors of house prices, providing a solid foundation for further modeling and insights into the factors driving these expenses.

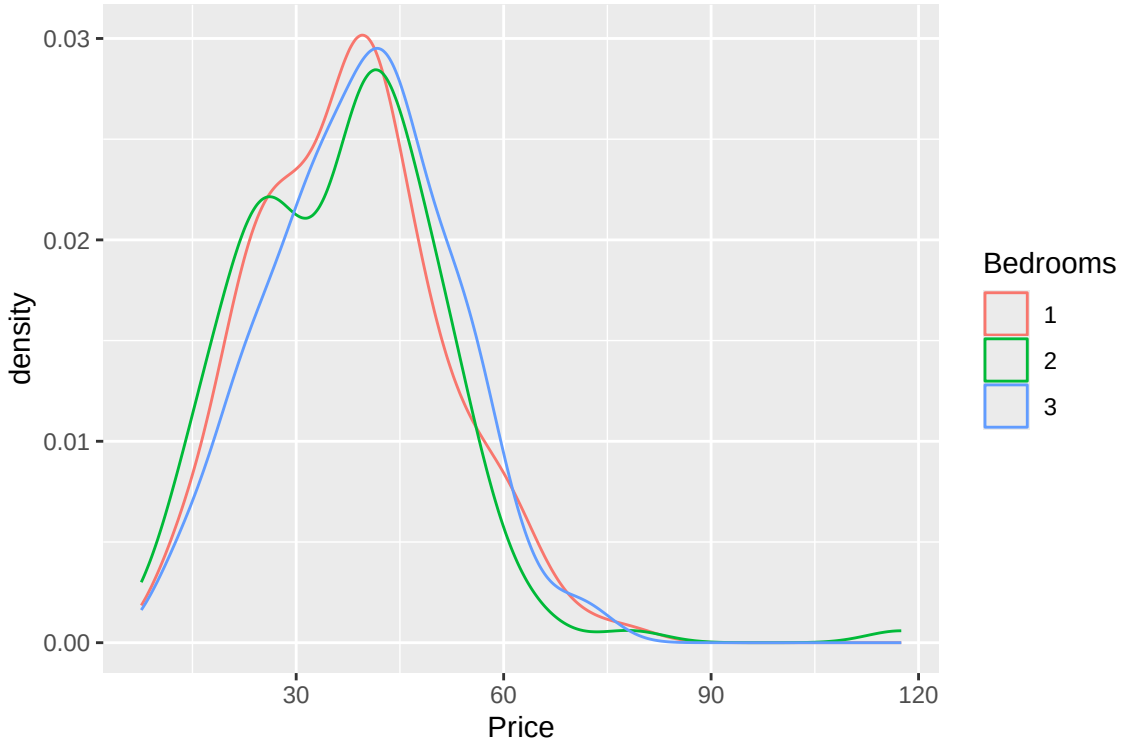


The density plot of the target variable shows some key observations:

- Positive (or right) skewness: The distribution is positively skewed, with the majority of data concentrated in the first part of the distribution. This indicates that most houses have moderate prices for unit area.
- Long Tail: The plot shows a long tail extending towards higher prices, which represents the presence of low frequency of high price values.

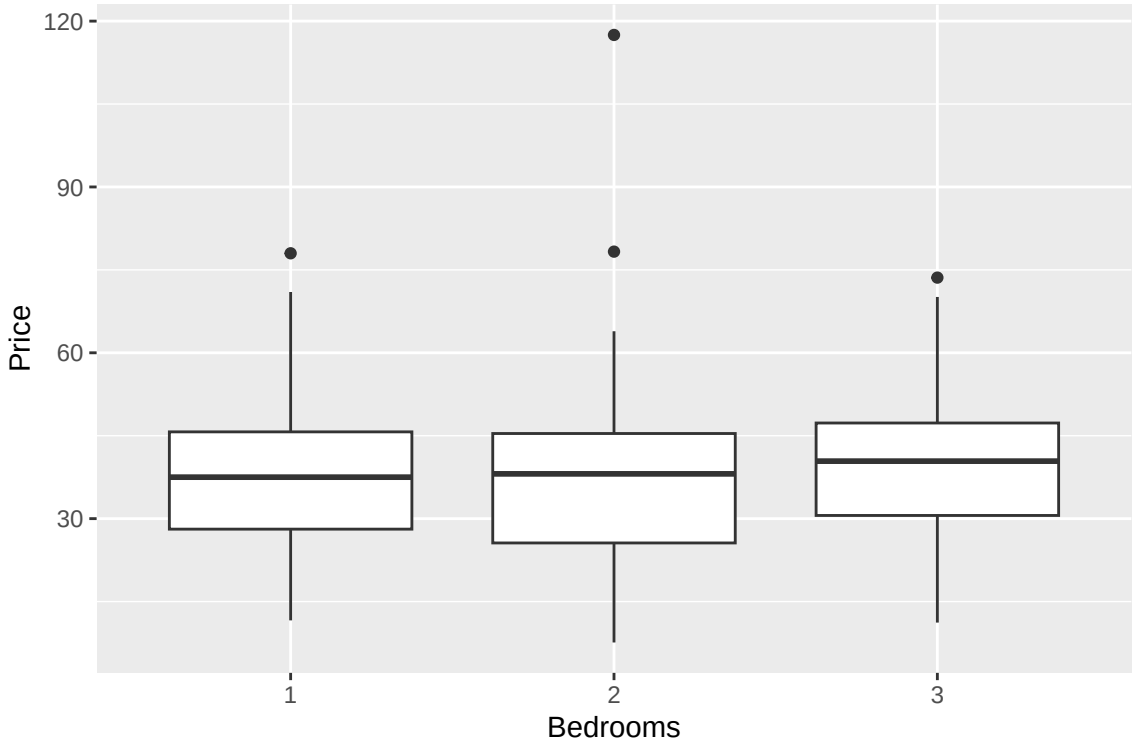
Now let's see how the price of houses varies in relation to some other variables of the dataset:



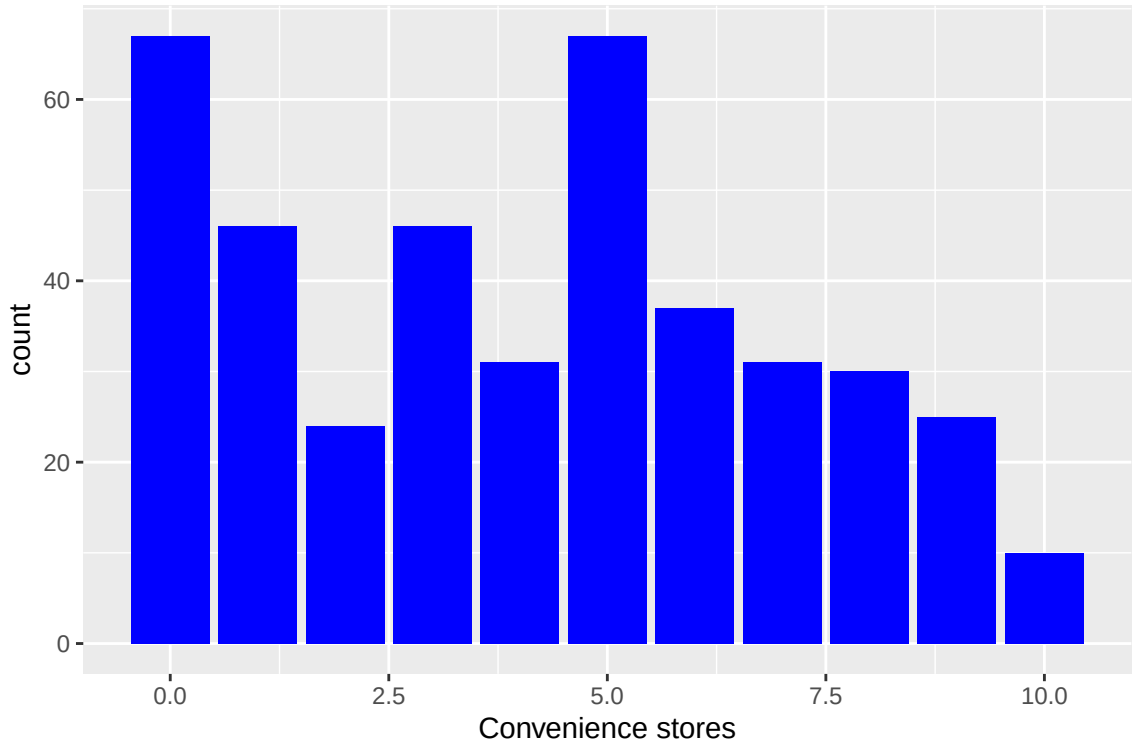


The bar chart related to the number of rooms shows that we have a strong balance within the dataset (houses with 1, 2, or 3 rooms are almost equal in number). Observing the frequency curves related to the three values of bedrooms, we notice that low prices are slightly more frequent for houses with only 1 room,

while, as we might have expected, the frequency of houses with higher prices is somewhat higher for houses with the largest number of rooms.

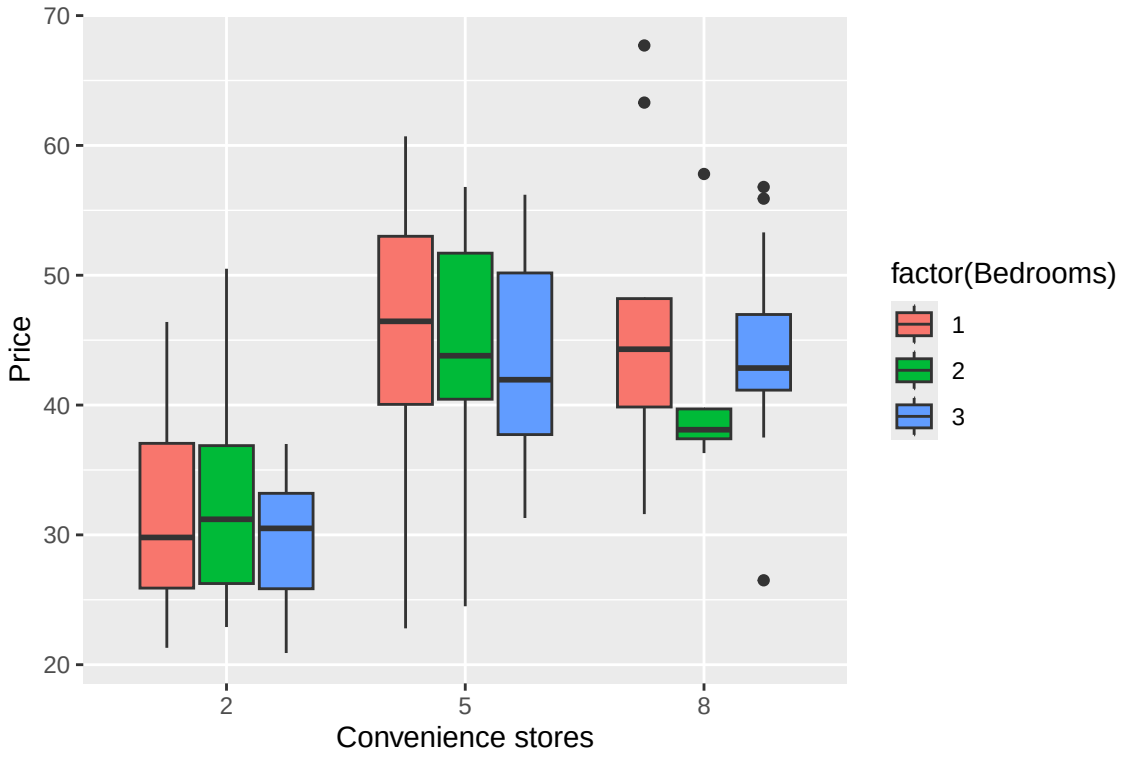


The boxplot once again highlights what has just seen.

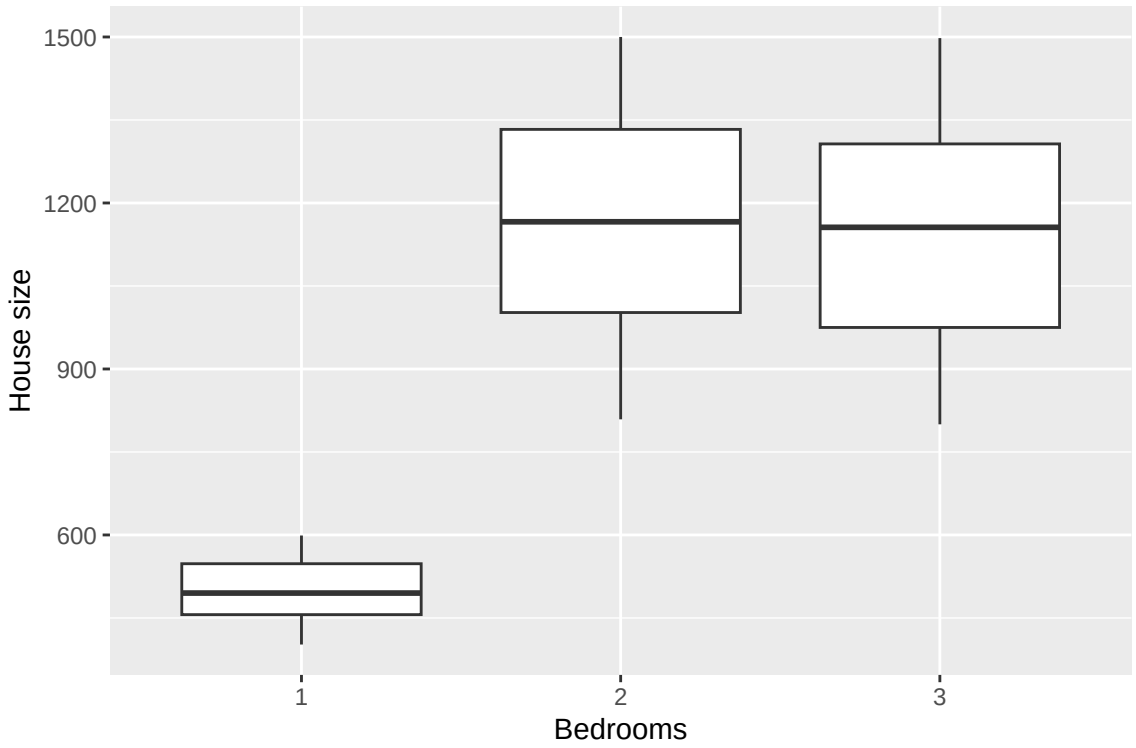


The variable Convenience stores shows that we have more households that have 5 stores nearby or households that have none at all. In order to have an accurate analysis that is not overly redundant, we will visualize only three levels of the number of stores in proximity in relation to the variable price.

Let's see how the price behaves in relation to the number of rooms and the number of stores near the house

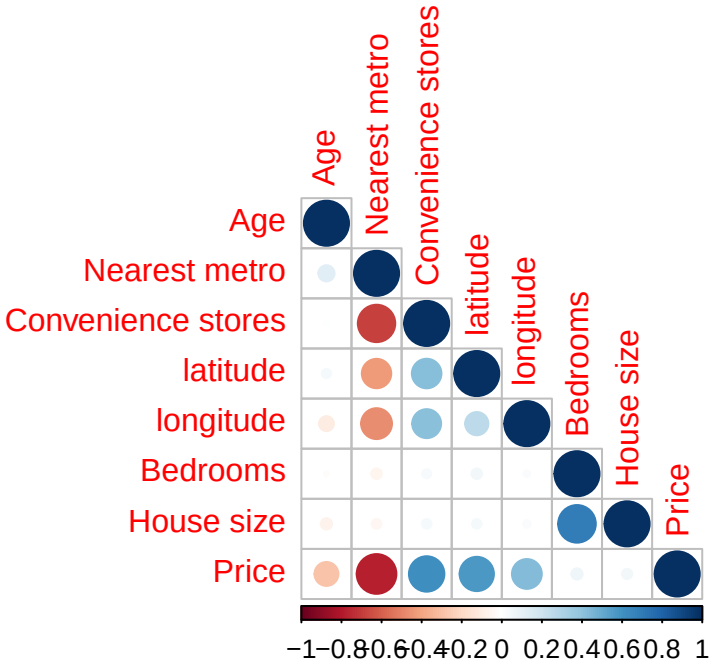


The price of houses increases significantly when moving from 2 to 5 stores nearby, while for houses that have between 5 to 8 stores close by, the price value does not seem to change much. What is rather strange is that when there are more stores nearby, houses with more rooms cost less; this could be because there may be other factors affecting that or because having many stores in the vicinity makes the number of rooms less influential.



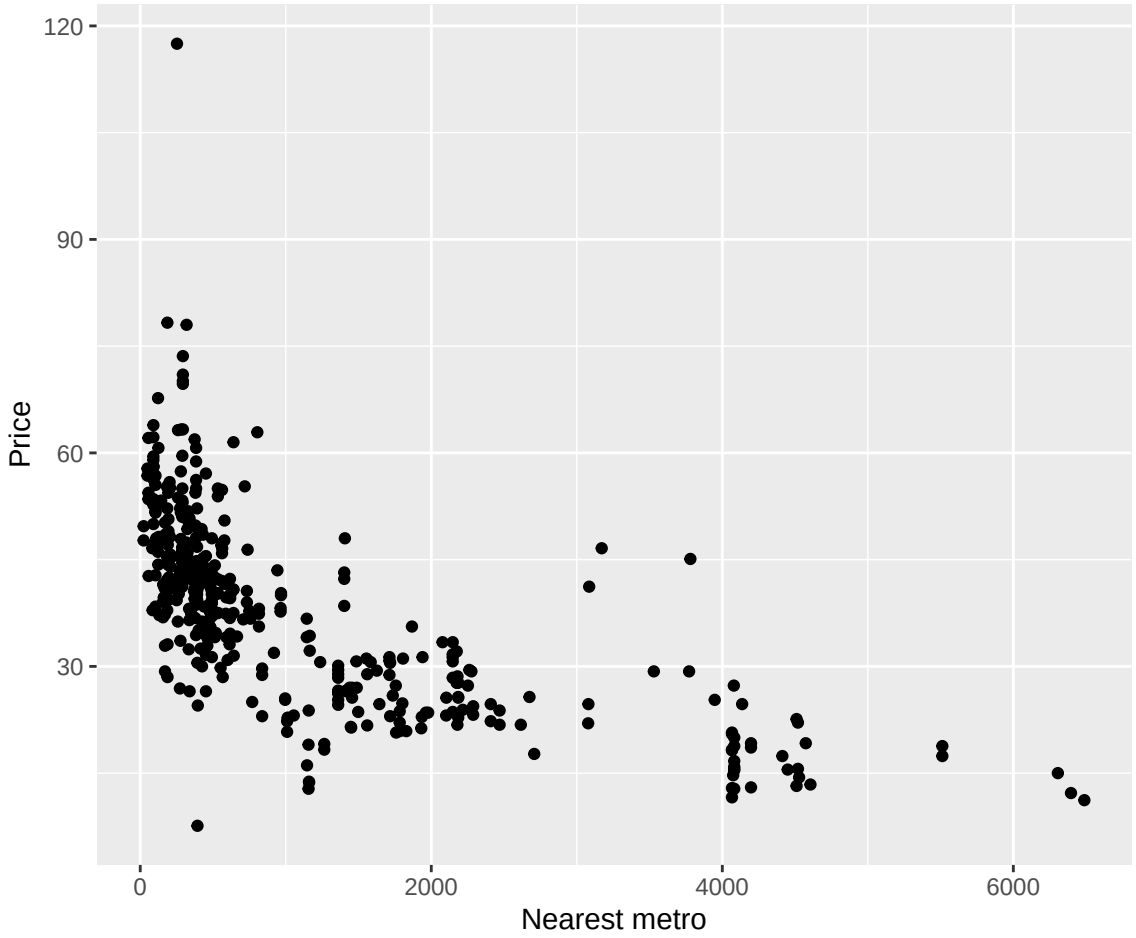
Even the size of the house does not increase much when going from 2 to 3 rooms. This could further demonstrate that the number of rooms does not have such a strong influence on the price of houses.

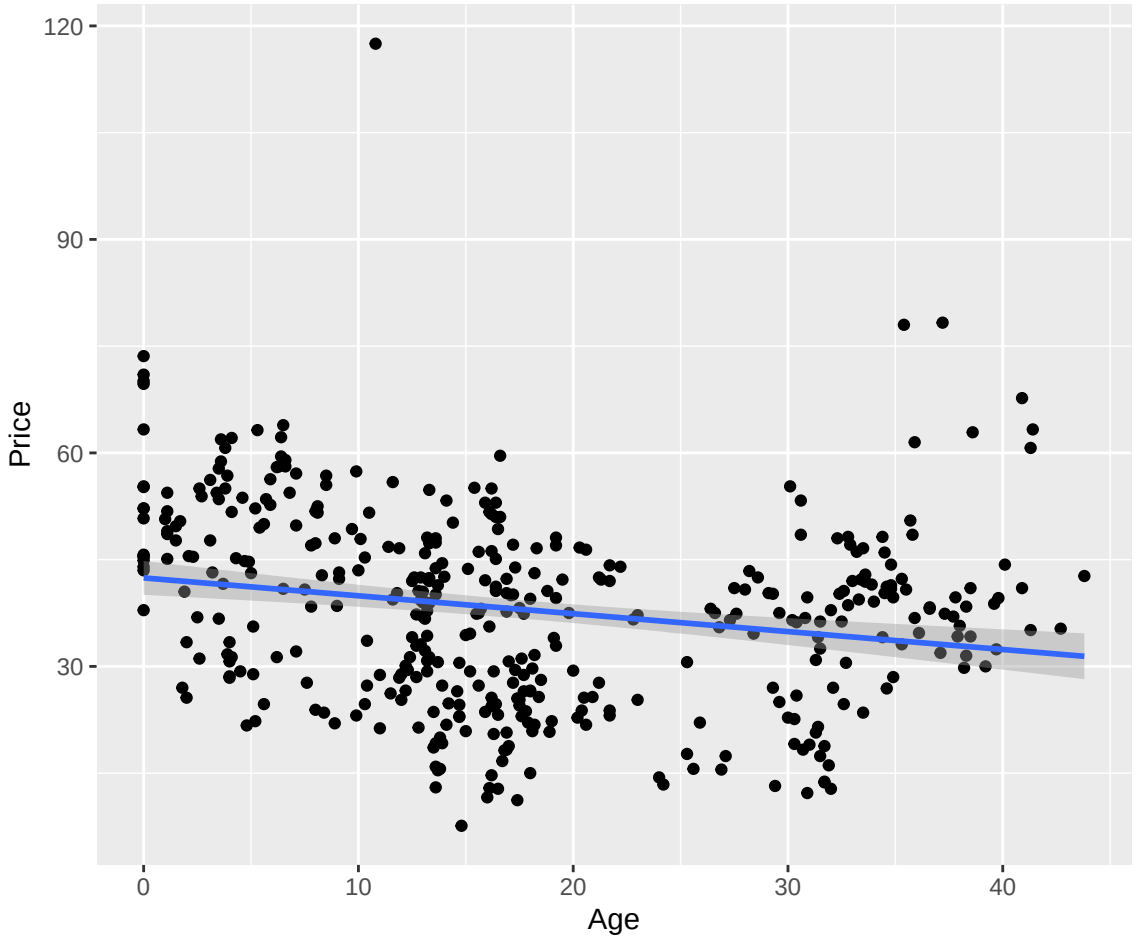
we'll now look at the correlation matrix, in order to uncover the existence of linear relationships between the continuous variables.

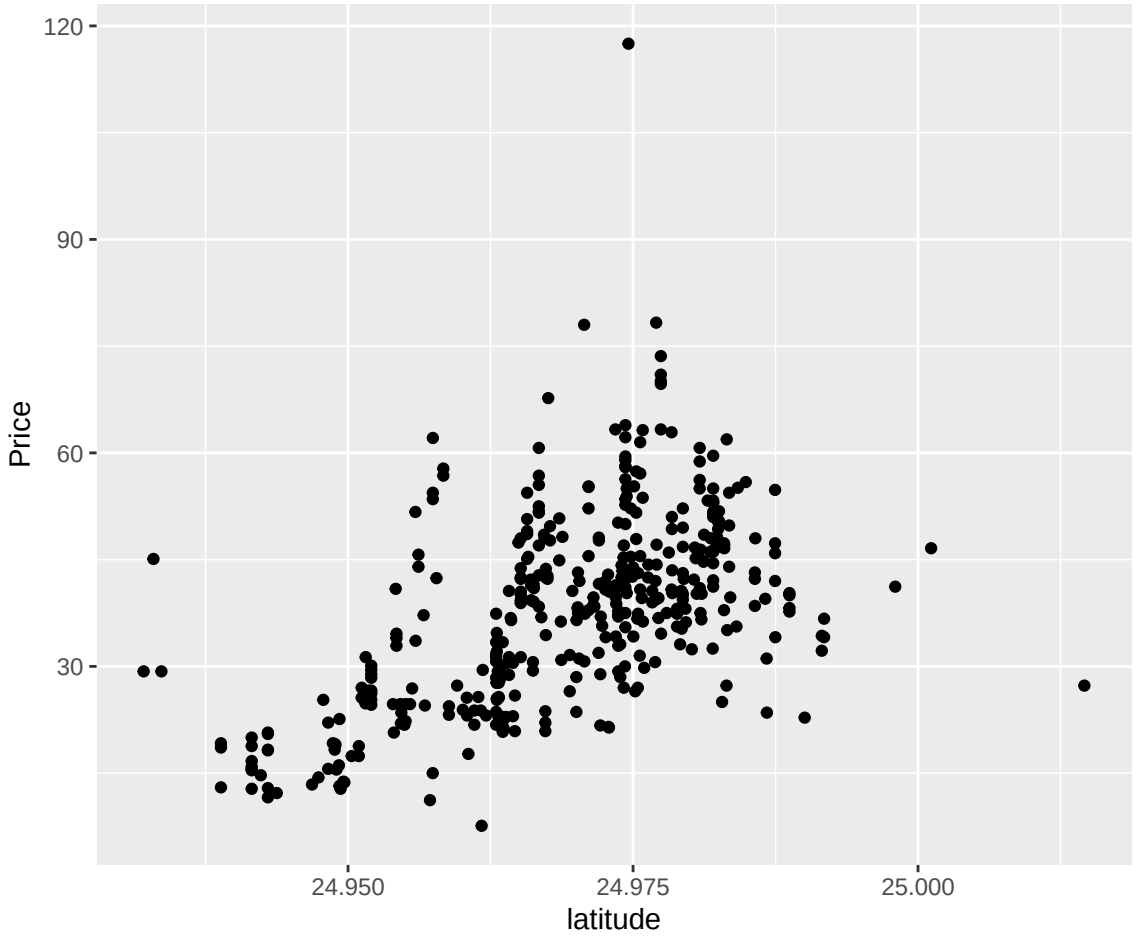


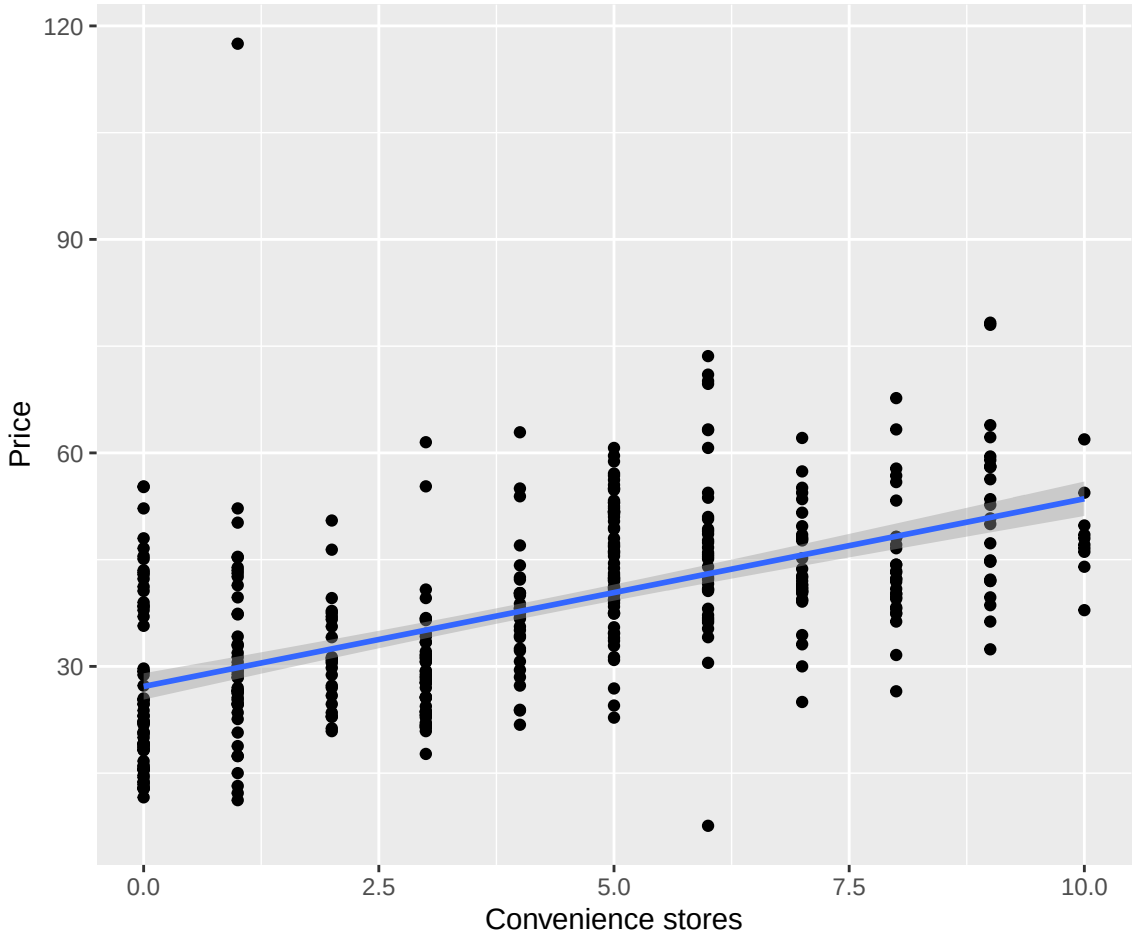
The correlation matrix shows some important results:

- The price is negatively correlated with the age of the house (weakly) and rather strongly with the distance from the nearest metro, while it has a positive linear relationship with the number of stores nearby, with latitude, and with longitude. The relationship with the number of rooms is close to zero.
- The size of the houses increases with the number of rooms.
- The farther the metro is, the fewer stores are near the house.


120
90
60
30
Price
0
2000
4000
6000
Nearest metro


120
90
60
30
Price
0
10
20
30
40
Age


120
90
60
30
Price
24.950
24.975
25.000
latitude



The scatter plots displayed graphically show some results obtained from the correlation matrix.

- The farther the nearest station is, the lower the price of the house will be.
- The price decreases as the age of the house increases.
- The price increases as the degrees of latitude increase. This could be due to the fact that as latitude increases, the climate tends to be milder and therefore more comfortable to consider living in that place.
- As already seen, the price increases with the increase in the number of nearby stores.

2. Model estimation

Before proceeding with the model estimation, to confirm that the target variable follows a skewed distribution, let's visualize the results of the Shapiro-Wilk test.

```
##
##  Shapiro-Wilk normality test
##
## data:  ff$Price
## W = 0.97275, p-value = 5.411e-07
```

This test is also called a normality test; under H0 we have the null hypothesis of normality. The value highly close to zero suggests that we reject this hypothesis, and consider the distribution of the variable as non-normal.

The model

Now, we focus on the application of a Gamma regression model to predict house prices. The Gamma regression is particularly suited for this analysis, as the target variable is continuous, positive, and exhibits a positively skewed distribution, making it a natural fit for the Gamma distribution within the Generalized Linear Model framework.

To ensure robust evaluation and reliable results, the dataset was divided into two subsets:

- Training set (80%): Used to train the model and identify the best predictors of the target variable.
- Test set (20%): Reserved for evaluating the model's predictive performance on unseen data.

Through a systematic model selection process, we identified the best-performing model, a Gamma regression with canonical link (inverse function).

$$E(Y|X) = \frac{1}{x'\beta} \quad con \quad Y|x \sim Gamma(\mu, \alpha)$$

the variables included in the model, useful for studying the target variable, are as follows:

- Age
- Nearest metro
- Convenience stores
- latitude

The model can be written this way:

$$E(Price) = \frac{1}{\beta_0 + Age\beta_1 + Nearestmetro\beta_2 + Conveniencestores\beta_3 + latitude\beta_4}$$

where β_0 represents the intercept, that is the expected value of the response variable when all the predictors are equal to zero, while while the other β_s represent the effect of each individual predictor variable on the response variable.

```
##
## Call:
## glm(formula = Price ~ Age + `Nearest metro` + `Convenience stores` +
##     latitude, family = Gamma(link = inverse), data = train_set)
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            5.882e+00  8.795e-01   6.688 9.84e-11 ***
## Age                    1.527e-04  2.658e-05   5.745 2.10e-08 ***
## `Nearest metro`        6.275e-06  5.221e-07  12.018  < 2e-16 ***
## `Convenience stores` -4.963e-04  1.310e-04  -3.789  0.00018 ***
## latitude             -2.347e-01  3.522e-02  -6.664 1.14e-10 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.0500953)
##
##     Null deviance: 46.280  on 330  degrees of freedom
## Residual deviance: 15.465  on 326  degrees of freedom
## AIC: 2301.1
##
## Number of Fisher Scoring iterations: 4
```

Although the coefficients are difficult to interpret, the best model for predicting prices has turned out to be this one. In fact, being an additive model (not multiplicative as in the case where link=log), it is not possible to obtain the actual magnitude of the predictors, since every variation is a function of η, and therefore of all the other predictors.

The only thing we can say by looking at the estimates is that negative values have a positive impact on the mean, and positive estimates have a negative impact on it.

In this case, the estimated model is:

$$E(Price) = \frac{1}{5.882 + 1.527 * 10^{-4} Age + 6.275 * 10^{-6} Nearestmetro - 4.963 * 10^{-4} Conveniencestores - 0.2347 latitude}$$

The model shows a good fit to the data, with all predictors being statistically significant.

Let's see a brief interpretation of the coefficients:

- Intercept: represents the baseline value for the price when all other coefficients are set to zero or at their reference.
- Age: the estimate value of Age is a positive value, so we can say that a unit increase in age results in an average decrease in price.
- Nearest metro: like the previous predictor, the β value is positive. The increase of 1 km from the nearest metro results in an average decrease in price.
- Convenience stores: We have a negative β value; this means that as the number of nearby stores increases, the average price of houses tends to rise.
- latitude: also in this case, a unit increase in latitude raises the average price of housing.

The Model Fit Statistics give the following results:

- Null deviance: 46.28 (degrees of freedom = 330), which represents the deviance of a model with no predictors (null model).
- Residual deviance: 15.465 (degrees of freedom = 326), indicating a substantial reduction in deviance with the inclusion of predictors. This suggests the model explains much of the variability in the data.
- AIC: 2301: A lower AIC value indicates better model fit, though it's best compared with other candidate models. Among the models compared, this one has the lowest AIC value.

The estimated dispersion parameter ϕ for the Gamma family is 0.05, indicating the level of variability in the data relative to the mean.

Marginal effects

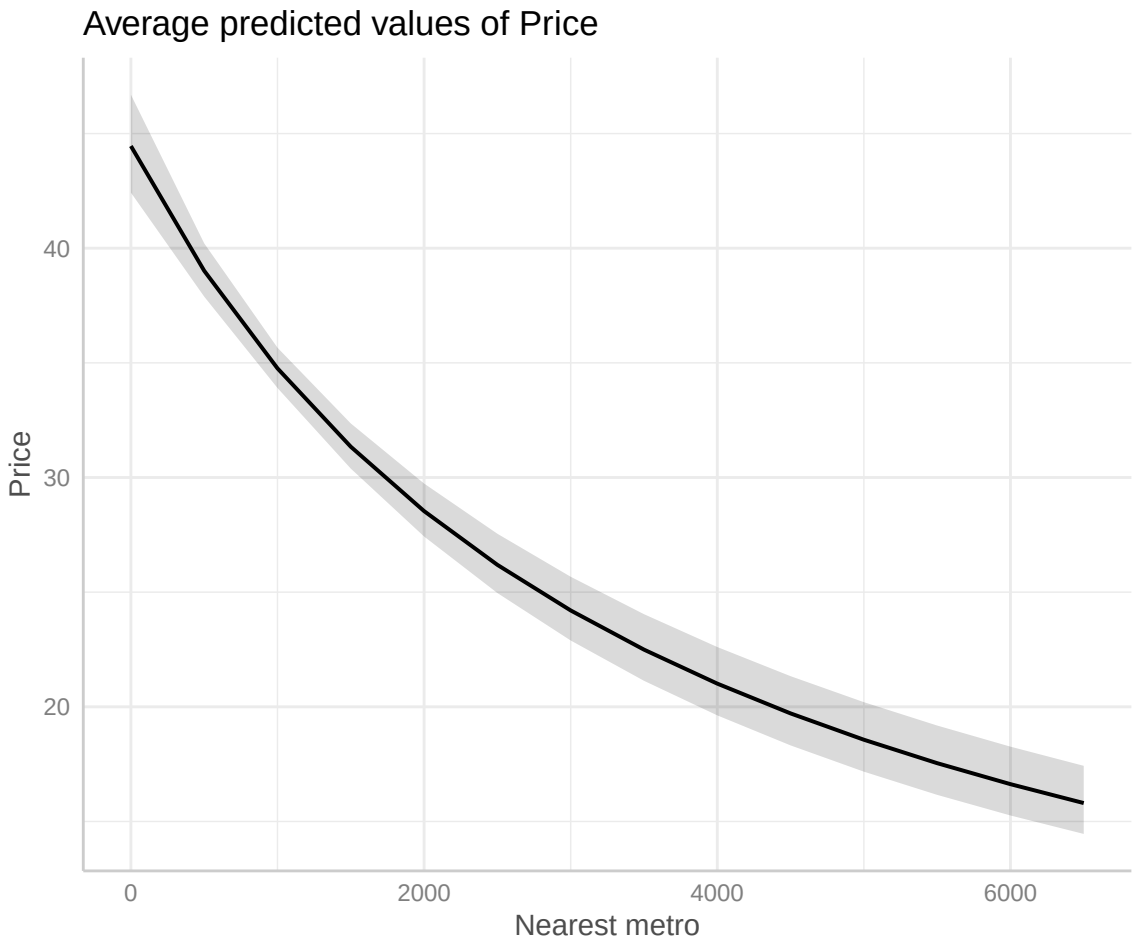
As already said in the previous paragraph, given the nature of the coefficients of the model with inverse link function, it's not possible to have an exact interpretation of these latter ones. For this reason, it's good norm to observe the marginal effects, which are true forecasts that allow us to have an intuitive idea of the impact of the predictive variables on the output.

$$\frac{\partial E(Price)}{\partial X_j} = -\frac{\beta_j}{x'\beta^2}$$

They answer the question: "What is the average variation of the response variable, in units or as a percentage, for a small change in a predictive variable, holding the others constant?".

Here are depicted the marginal effects related to the variables in question:

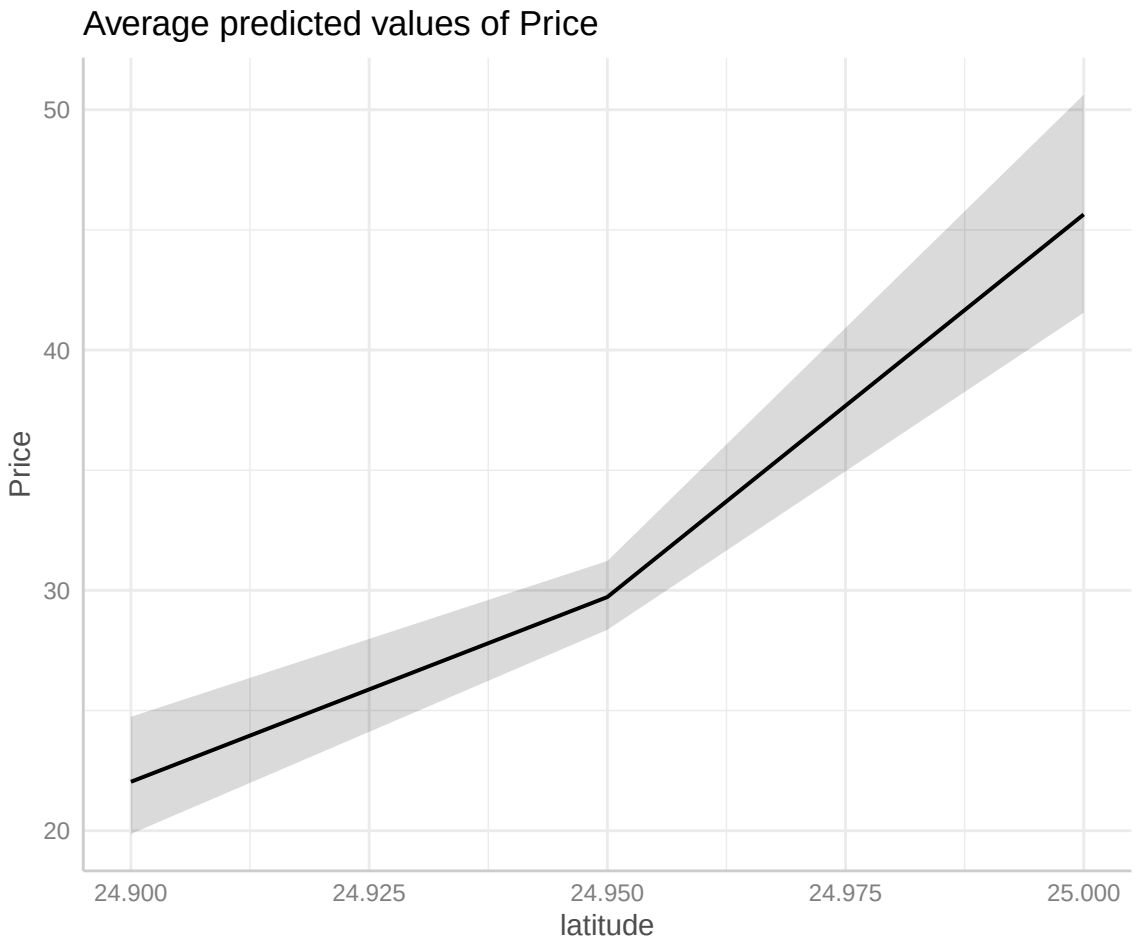
```
## # Average predicted values of Price
##
## Nearest metro | Predicted |       95% CI
## ---------------------------------------
##             0 |     44.46 | 42.42, 46.69
##          1000 |     34.76 | 33.90, 35.66
##          1500 |     31.34 | 30.39, 32.36
##          2500 |     26.19 | 24.96, 27.55
##          3500 |     22.49 | 21.13, 24.04
##          4000 |     21.01 | 19.63, 22.61
##          5000 |     18.56 | 17.17, 20.20
##          6500 |     15.80 | 14.46, 17.43
```



First plot shows the marginal effects of the nearest metro variable on house prices. It can be noticed, as we had already seen from the model, a decreasing effect of the price as the distance in kilometers from the metro station increases; effect that at first seems to be stronger, given the greater slope of the curve, only to gradually start to ease off.

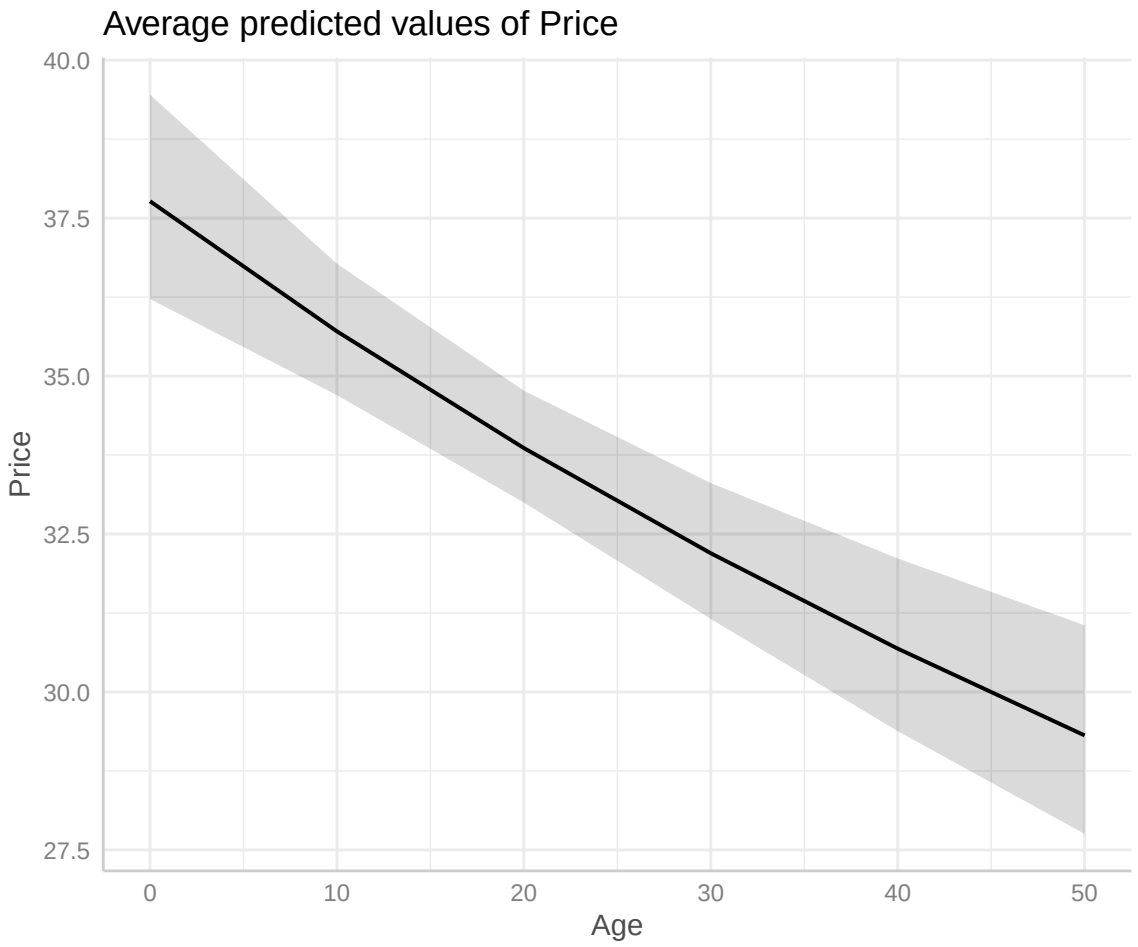
The gray area around the curve represents the forecast interval; being quite narrow, it shows a certain accuracy in the prediction.

```
## # Average predicted values of Price
##
## latitude | Predicted |       95% CI
## -----------------------------------
##    24.90 |     22.04 | 19.87, 24.74
##    24.95 |     29.72 | 28.36, 31.22
##    25.00 |     45.64 | 41.55, 50.63
```

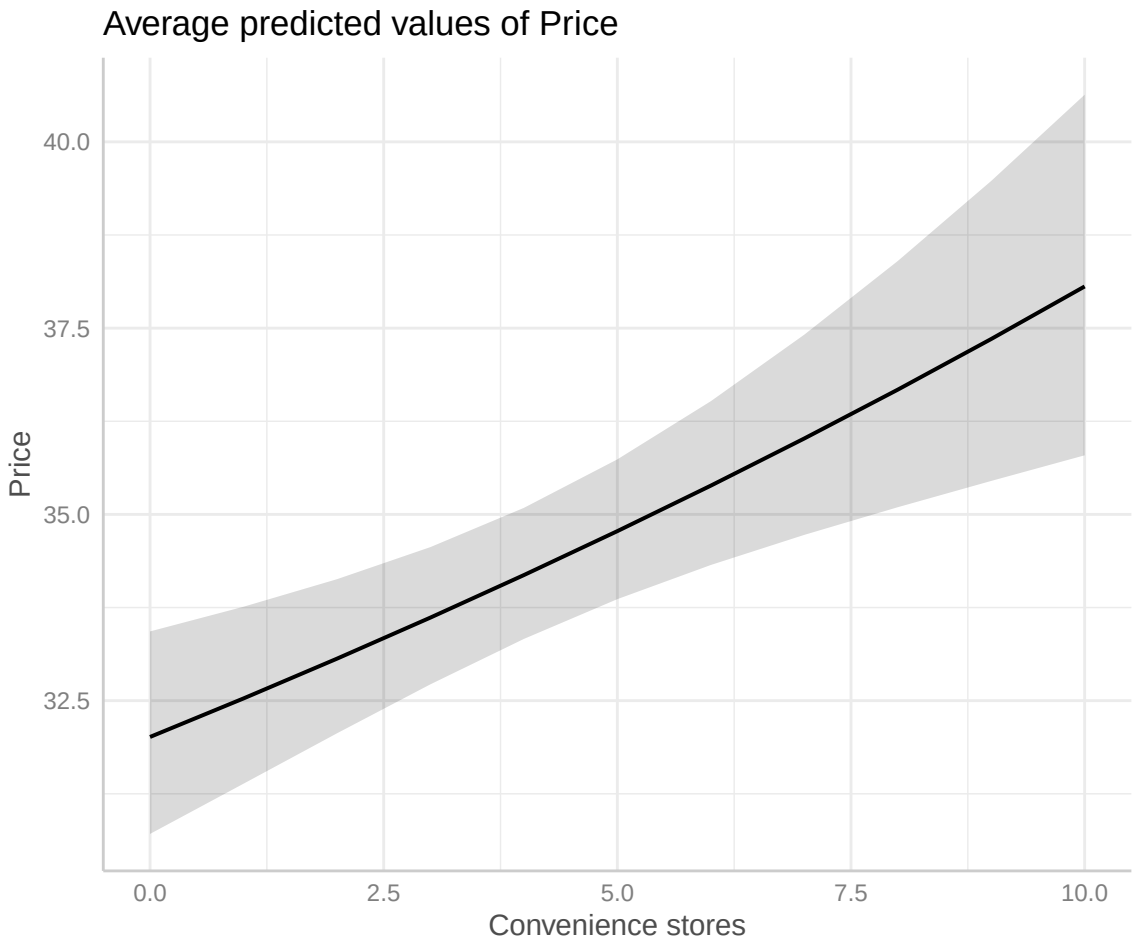


Marginal effect of latitude shows show a little more uncertainty in the predictions. We can also observe non-linearity here, but for this predictor, the effect on the price is increasing. Effect that increases for higher values of the predictor

```
## # Average predicted values of Price
##
## Age | Predicted |       95% CI
## ------------------------------
##   0 |     37.77 | 36.22, 39.45
##  10 |     35.71 | 34.70, 36.78
##  20 |     33.86 | 33.00, 34.77
##  30 |     32.20 | 31.16, 33.31
##  40 |     30.69 | 29.38, 32.11
##  50 |     29.31 | 27.75, 31.06
```



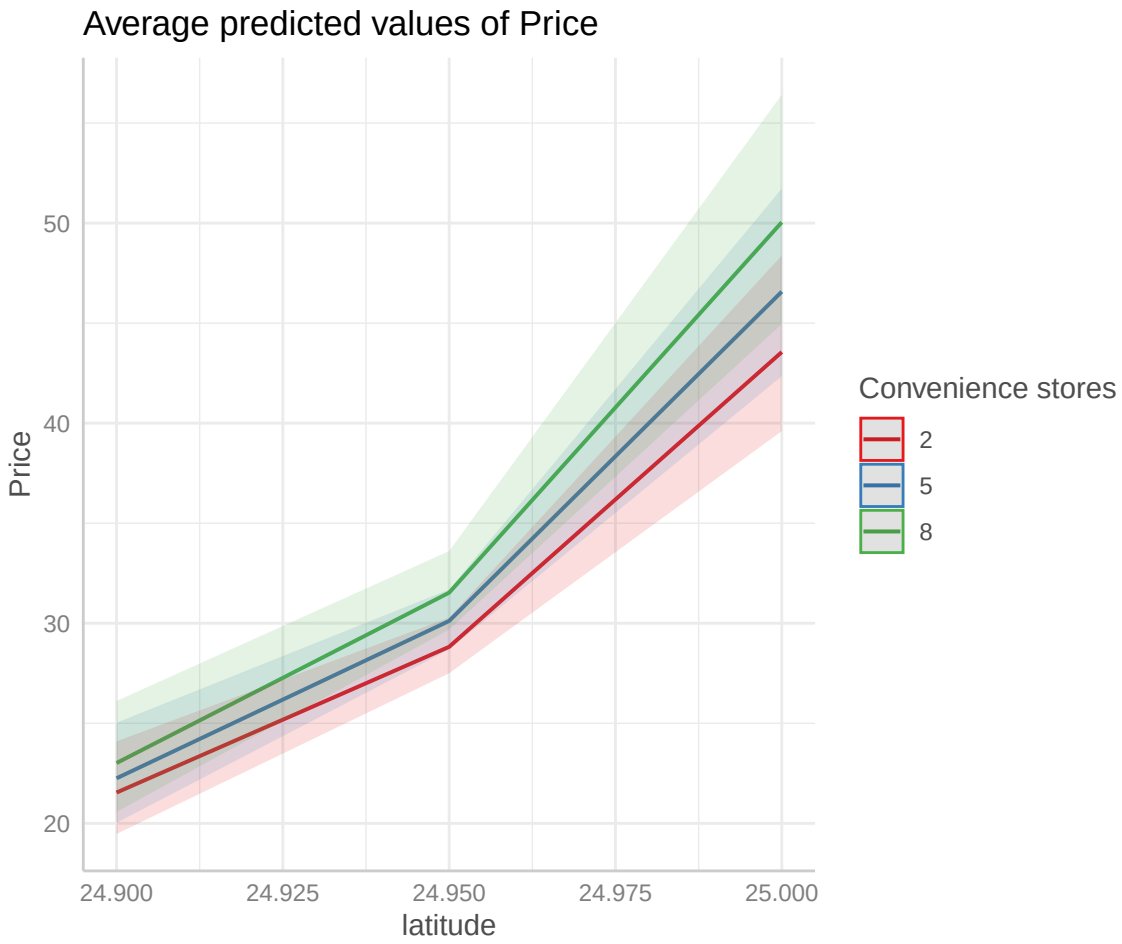

```
## # Average predicted values of Price
##
## Convenience stores | Predicted |       95% CI
## -----------------------------------------------
##                  0 |     32.01 | 30.71, 33.43
##                  1 |     32.53 | 31.39, 33.76
##                  3 |     33.62 | 32.72, 34.56
##                  4 |     34.19 | 33.33, 35.09
##                  5 |     34.78 | 33.86, 35.74
##                  6 |     35.39 | 34.32, 36.52
##                  7 |     36.02 | 34.72, 37.41
##                 10 |     38.06 | 35.79, 40.63
```



Finally, the last two predictive variables reflect once again what was observed in the estimates, with prediction intervals showing greater forecasting uncertainty compared to the first two.

Now let's move on to the observation of conditioned marginal effects, taking into account the convenience store variable as a conditioning factor. For simplicity, we will not visualize all levels of the count variable.


Average predicted values of Price
Price
50
40
30
20
0
2000
4000
6000
Nearest metro
Convenience stores
2
5
8


Average predicted values of Price
Price
50
40
30
20
24.900
24.925
24.950
24.975
25.000
latitude
Convenience stores
2
5
8



The situation is identical to the effects seen individually for each predictor, but this time three curves are represented, related to the number of stores near the house. As we could have expected, the conditioned marginal effect does not undermine the analysis done so far. The price is always positively correlated with convenience stores that are located near homes.

Predictions

We can conclude the analysis with the predictions.

Predictions that will be made on the sample that we excluded from the training (test set).

```
##     Price Predicted_values
## 9    18.8            15.39
## 11   41.4            35.56
## 28   33.6            38.88
## 35   55.1            54.30
## 43   34.7            33.90
## 54   38.9            39.44
## 67   50.7            47.67
## 70   42.0            48.94
## 76   29.5            28.00
## 77   36.8            36.04
```

```
## 84   17.7            23.17
## 93   21.8            24.80
## 97   59.5            56.69
## 98   34.6            37.66
## 99   51.0            48.56
## 109  34.1            35.10
## 111  51.6            45.96
## 121  31.3            39.44
## 132  30.7            27.92
## 134  39.5            53.18
```

In the table shown above, we observe a comparison between the values of the test set and those predicted by the model.

Although the model used is the most accurate for predictions, we note that these are not sufficiently precise.

However, the model predictions are often inaccurate. These values could be influenced by thepresence of outliers.

3. Conclusions

In conclusion, we can state that the analysis conducted has led to some results that will be listed here:

- with the model used it was not possible to assess the effective magnitude of the estimated coefficients, but thanks to the marginal effects we can get an idea about it. It can indeed be observed from these effects that the variables with the greatest influence on housing prices seem to be latitude and distance from the nearest metro.
- The other 2 variables show moderate but significant effects; the variable Age reflects how a newer house has a higher price. 'Convenience store' shows that having greater comfort and speed in grocery shopping leads to an increase in the average price of a home.

Results that can be useful both from the perspective of buyers and that of property owners:

- For homeowners, knowing the key factors that influence price can be a great competitive advantage.
- For buyers, Having this information is important because it allows for making the right choice considering the quality-price ratio.

4. Appendix

Model selection

Inside this paragraph, the procedures for selecting the model used will be shown.

In addition to the 'inverse' activation link, models with non-canonical links have also been tested:

Gamma regression with link='log':

```
##
## Call:
## glm(formula = Price ~ ., family = Gamma(link = log), data = train_set)
```

```
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -3.439e+02  1.918e+02  -1.793   0.0739 .
## Age                  -6.130e-03  1.125e-03  -5.449 1.00e-07 ***
## `Nearest metro`      -1.264e-04  2.174e-05  -5.814 1.47e-08 ***
## `Convenience stores`  3.115e-02  5.489e-03   5.675 3.08e-08 ***
## latitude              7.903e+00  1.287e+00   6.141 2.41e-09 ***
## longitude             1.237e+00  1.533e+00   0.807   0.4203
## Bedrooms              2.042e-02  2.384e-02   0.856   0.3924
## `House size`         -4.337e-05  5.685e-05  -0.763   0.4460
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.05383436)
##
##     Null deviance: 46.280  on 330  degrees of freedom
## Residual deviance: 16.305  on 323  degrees of freedom
## AIC: 2324.8
##
## Number of Fisher Scoring iterations: 5
```

```
##
## Call:
## glm(formula = Price ~ Age + `Nearest metro` + `Convenience stores` +
##     latitude, family = Gamma(link = log), data = train_set)
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.913e+02  3.195e+01  -5.989 5.58e-09 ***
## Age                  -6.077e-03  1.121e-03  -5.420 1.16e-07 ***
## `Nearest metro`      -1.401e-04  1.420e-05  -9.866  < 2e-16 ***
## `Convenience stores`  3.079e-02  5.462e-03   5.637 3.76e-08 ***
## latitude              7.812e+00  1.279e+00   6.106 2.91e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.0537414)
##
##     Null deviance: 46.280  on 330  degrees of freedom
## Residual deviance: 16.374  on 326  degrees of freedom
## AIC: 2320.2
##
## Number of Fisher Scoring iterations: 5
```

Gamma regression with link='identity':

```
##
## Call:
## glm(formula = Price ~ ., family = Gamma(link = inverse), data = train_set)
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)           9.718e+00  5.786e+00   1.680 0.094000 .
## Age                   1.549e-04  2.678e-05   5.782 1.74e-08 ***
## 'Nearest metro'       5.969e-06  6.887e-07   8.666  < 2e-16 ***
## 'Convenience stores' -4.958e-04  1.323e-04  -3.747 0.000212 ***
## latitude             -2.353e-01  3.533e-02  -6.659 1.19e-10 ***
## longitude            -3.145e-02  4.702e-02  -0.669 0.504069
## Bedrooms             -3.964e-04  5.997e-04  -0.661 0.509081
## 'House size'          1.245e-06  1.440e-06   0.865 0.387733
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.05038647)
##
##     Null deviance: 46.280  on 330  degrees of freedom
## Residual deviance: 15.408  on 323  degrees of freedom
## AIC: 2305.9
##
## Number of Fisher Scoring iterations: 4
```

```
##
## Call:
## glm(formula = Price ~ Age + 'Nearest metro' + 'Convenience stores' +
##     latitude, family = Gamma(link = inverse), data = train_set)
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           5.882e+00  8.795e-01   6.688 9.84e-11 ***
## Age                   1.527e-04  2.658e-05   5.745 2.10e-08 ***
## 'Nearest metro'       6.275e-06  5.221e-07  12.018  < 2e-16 ***
## 'Convenience stores' -4.963e-04  1.310e-04  -3.789  0.00018 ***
## latitude             -2.347e-01  3.522e-02  -6.664 1.14e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.0500953)
##
##     Null deviance: 46.280  on 330  degrees of freedom
## Residual deviance: 15.465  on 326  degrees of freedom
## AIC: 2301.1
##
## Number of Fisher Scoring iterations: 4
```

we see that these models also present the same significant variables; for this reason, the comparison will be carried out directly on the reduced models.

Let's now look at models that differ not only in the link function but also in the distribution function used.

The other function that can be used in cases of positive asymmetry is the 'inverse Gaussian' regression.

IG regression with link='inverse':

```
##
## Call:
## glm(formula = Price ~ ., family = inverse.gaussian(link = inverse),
##     data = train_set)
```

```
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.009e+01  6.123e+00   1.648 0.100427
## Age                    1.559e-04  3.179e-05   4.904 1.49e-06 ***
## `Nearest metro`        5.479e-06  7.053e-07   7.768 1.07e-13 ***
## `Convenience stores` -5.949e-04  1.558e-04  -3.817 0.000162 ***
## latitude             -2.585e-01  3.918e-02  -6.597 1.72e-10 ***
## longitude            -2.971e-02  4.937e-02  -0.602 0.547724
## Bedrooms             -5.843e-04  6.991e-04  -0.836 0.403867
## `House size`          1.571e-06  1.676e-06   0.937 0.349531
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.001713731)
##
##     Null deviance: 1.47334  on 330  degrees of freedom
## Residual deviance: 0.53612  on 323  degrees of freedom
## AIC: 2372.4
##
## Number of Fisher Scoring iterations: 2
```

```
##
## Call:
## glm(formula = Price ~ Age + `Nearest metro` + `Convenience stores` +
##     latitude, family = inverse.gaussian(link = inverse), data = train_set)
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            6.441e+00  9.762e-01   6.599 1.68e-10 ***
## Age                    1.533e-04  3.165e-05   4.845 1.96e-06 ***
## `Nearest metro`        5.792e-06  4.959e-07  11.680  < 2e-16 ***
## `Convenience stores` -5.899e-04  1.552e-04  -3.802 0.000171 ***
## latitude             -2.571e-01  3.909e-02  -6.576 1.92e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.001710876)
##
##     Null deviance: 1.47334  on 330  degrees of freedom
## Residual deviance: 0.53811  on 326  degrees of freedom
## AIC: 2367.6
##
## Number of Fisher Scoring iterations: 2
```

Let's check an IG regression model with link='log':

```
##
## Call:
## glm(formula = Price ~ ., family = inverse.gaussian(link = log),
##     data = train_set)
##
## Coefficients:
```

```
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -3.688e+02  1.977e+02  -1.866   0.0629 .
## Age                  -5.789e-03  1.271e-03  -4.556 7.39e-06 ***
## `Nearest metro`      -1.106e-04  2.185e-05  -5.062 6.97e-07 ***
## `Convenience stores`  3.428e-02  6.198e-03   5.531 6.57e-08 ***
## latitude              7.716e+00  1.360e+00   5.676 3.08e-08 ***
## longitude             1.480e+00  1.568e+00   0.944   0.3460
## Bedrooms              2.748e-02  2.540e-02   1.082   0.2800
## `House size`         -5.058e-05  5.986e-05  -0.845   0.3988
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.001822175)
##
##     Null deviance: 1.47334  on 330  degrees of freedom
## Residual deviance: 0.56416  on 323  degrees of freedom
## AIC: 2389.2
##
## Number of Fisher Scoring iterations: 7
```

```
##
## Call:
## glm(formula = Price ~ Age + `Nearest metro` + `Convenience stores` +
##     latitude, family = inverse.gaussian(link = log), data = train_set)
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.843e+02  3.375e+01  -5.460 9.44e-08 ***
## Age                  -5.766e-03  1.276e-03  -4.520 8.67e-06 ***
## `Nearest metro`      -1.279e-04  1.311e-05  -9.755  < 2e-16 ***
## `Convenience stores`  3.353e-02  6.192e-03   5.415 1.19e-07 ***
## latitude              7.529e+00  1.352e+00   5.570 5.35e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.001843599)
##
##     Null deviance: 1.47334  on 330  degrees of freedom
## Residual deviance: 0.56763  on 326  degrees of freedom
## AIC: 2385.3
##
## Number of Fisher Scoring iterations: 7
```

Finally, let's see how an Inverse Gaussian behaves with the identity function as the activation link.

```
##
## Call:
## glm(formula = Price ~ ., family = inverse.gaussian(link = identity),
##     data = train_set)
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.251e+04  6.569e+03  -1.904  0.05776 .
```

```
## Age                  -1.797e-01  4.417e-02  -4.068 5.96e-05 ***
## `Nearest metro`      -1.945e-03  6.603e-04  -2.946  0.00345 **
## `Convenience stores`  1.572e+00  2.283e-01   6.885 3.01e-11 ***
## latitude              2.011e+02  4.264e+01   4.718 3.55e-06 ***
## longitude             6.190e+01  5.182e+01   1.195  0.23312
## Bedrooms              9.598e-01  7.562e-01   1.269  0.20530
## `House size`         -1.632e-03  1.650e-03  -0.989  0.32327
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.001923541)
##
##     Null deviance: 1.4733  on 330  degrees of freedom
## Residual deviance: 0.6179  on 323  degrees of freedom
## AIC: 2419.4
##
## Number of Fisher Scoring iterations: 15

##
## Call:
## glm(formula = Price ~ Age + `Nearest metro` + `Convenience stores` +
##     latitude, family = inverse.gaussian(link = identity), data = train_set)
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -4.545e+03  1.039e+03  -4.375 1.63e-05 ***
## Age                  -1.818e-01  4.424e-02  -4.110 5.02e-05 ***
## `Nearest metro`      -2.742e-03  3.021e-04  -9.074  < 2e-16 ***
## `Convenience stores`  1.539e+00  2.287e-01   6.731 7.59e-11 ***
## latitude              1.835e+02  4.162e+01   4.410 1.41e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.001970752)
##
##     Null deviance: 1.4733  on 330  degrees of freedom
## Residual deviance: 0.6232  on 326  degrees of freedom
## AIC: 2416.2
##
## Number of Fisher Scoring iterations: 16
```

The statistically significant coefficients at the α level set at 0.05 are once again the same.

To compare the models, the AIC information criterion was used, which is useful for non-nested models.

```
##                df      AIC      BIC
## gamma_log2      6 2320.204 2343.017
## inverse_log2    6 2385.279 2408.092
## gamma_invers2   6 2301.137 2323.950
## invers_inv2     6 2367.602 2390.415
## gamma_id2       6 2301.137 2323.950
## invers_id2      6 2416.192 2439.005
```

The model with the lowest AIC and BIC values is the one chosen for the analysis.

To have further proof of the chosen model, we can perform a Likelihood-ratio test between the two Gamma models with 'inverse' link.

```
##
## Call:
## glm(formula = Price ~ ., family = Gamma(link = inverse), data = train_set)
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            9.718e+00  5.786e+00   1.680 0.094000 .
## Age                    1.549e-04  2.678e-05   5.782 1.74e-08 ***
## 'Nearest metro'        5.969e-06  6.887e-07   8.666  < 2e-16 ***
## 'Convenience stores' -4.958e-04  1.323e-04  -3.747 0.000212 ***
## latitude             -2.353e-01  3.533e-02  -6.659 1.19e-10 ***
## longitude            -3.145e-02  4.702e-02  -0.669 0.504069
## Bedrooms             -3.964e-04  5.997e-04  -0.661 0.509081
## 'House size'          1.245e-06  1.440e-06   0.865 0.387733
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.05038647)
##
##     Null deviance: 46.280  on 330  degrees of freedom
## Residual deviance: 15.408  on 323  degrees of freedom
## AIC: 2305.9
##
## Number of Fisher Scoring iterations: 4
```

```
## Analysis of Deviance Table
##
## Model 1: Price ~ Age + 'Nearest metro' + 'Convenience stores' + latitude +
##     longitude + Bedrooms + 'House size'
## Model 2: Price ~ Age + 'Nearest metro' + 'Convenience stores' + latitude
##   Resid. Df Resid. Dev Df  Deviance Pr(>Chi)
## 1       323     15.408
## 2       326     15.465 -3 -0.056535   0.7718
```

It makes sense to carry out this test when we have nested models. In the case under analysis, the value of the p-value strongly greater than the set level suggests that we should not reject the null hypothesis H0, according to which the two models have no differences. Therefore, by the principle of parsimony, the reduced model is chosen.

Finally, let's evaluate, through specific metrics, the predictive ability of the models just seen. The methods used will be the Mean Squared Error (MSE) and the Mean Absolute Error (MAE).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \qquad MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

```
##                    MSE      MAE
## gamma_log2  110.52221 5.706154
## invers_log2 113.40341 5.806968
## gamma_inv2   98.77853 4.984359
## invers_inv2 102.16430 5.087777
```

```
## gamma_id2    98.77853 4.984359
## invers_id2  127.53787 6.628304
```

As for the criteria of information, for these metrics as well, the model with the lowest values is the one with the best predictive accuracy. Once again we see that the model to choose is the Gamma regression with an inverse link.